

BUILT WITH CLAUDE SCIENCE · LIFE SCIENCES RESEARCH TRACK

From perturbation to a druggable target — in a week

Causal target discovery in primary human T cells.

Which genes, when perturbed, causally reshape a T cell — and which of those are real, druggable targets?

E-distance ranking · Human-genetics concordance · Reproducible provenance

HOW CLAUDE SCIENCE GOT US THERE

Reproducible by construction

Every result carries its code, its environment, and a plain-language trail.

● Code

Every figure traces to the exact script that made it.

● Environment

Locked dependencies — one command reproduces the run.

● Plain-language trail

A readable record of every step and decision.



A reviewer agent checks each claim against what actually ran.

It flagged an inflated statistic in our effect-size metric — so we caught and fixed a real bug in the energy-distance estimator before it ever reached a figure.

Null E-distance: ~5.0 → ~0.0 (bias removed, n-invariant)

THE METHOD

Rank by causal effect size, not p-value

With enough cells, everything is significant. So we rank on the magnitude of the causal shift — power-equalized energy distance — and use the permutation test only as a gate.

01

Energy distance

Rank on effect size, not statistical significance.

02

Viability + on-target

Gate out toxic knockouts and mis-edits.

03

Donors as replicates

Signal must hold across independent donors.

04

Genetics direction

Human-genetics direction-of-effect as the differentiator.

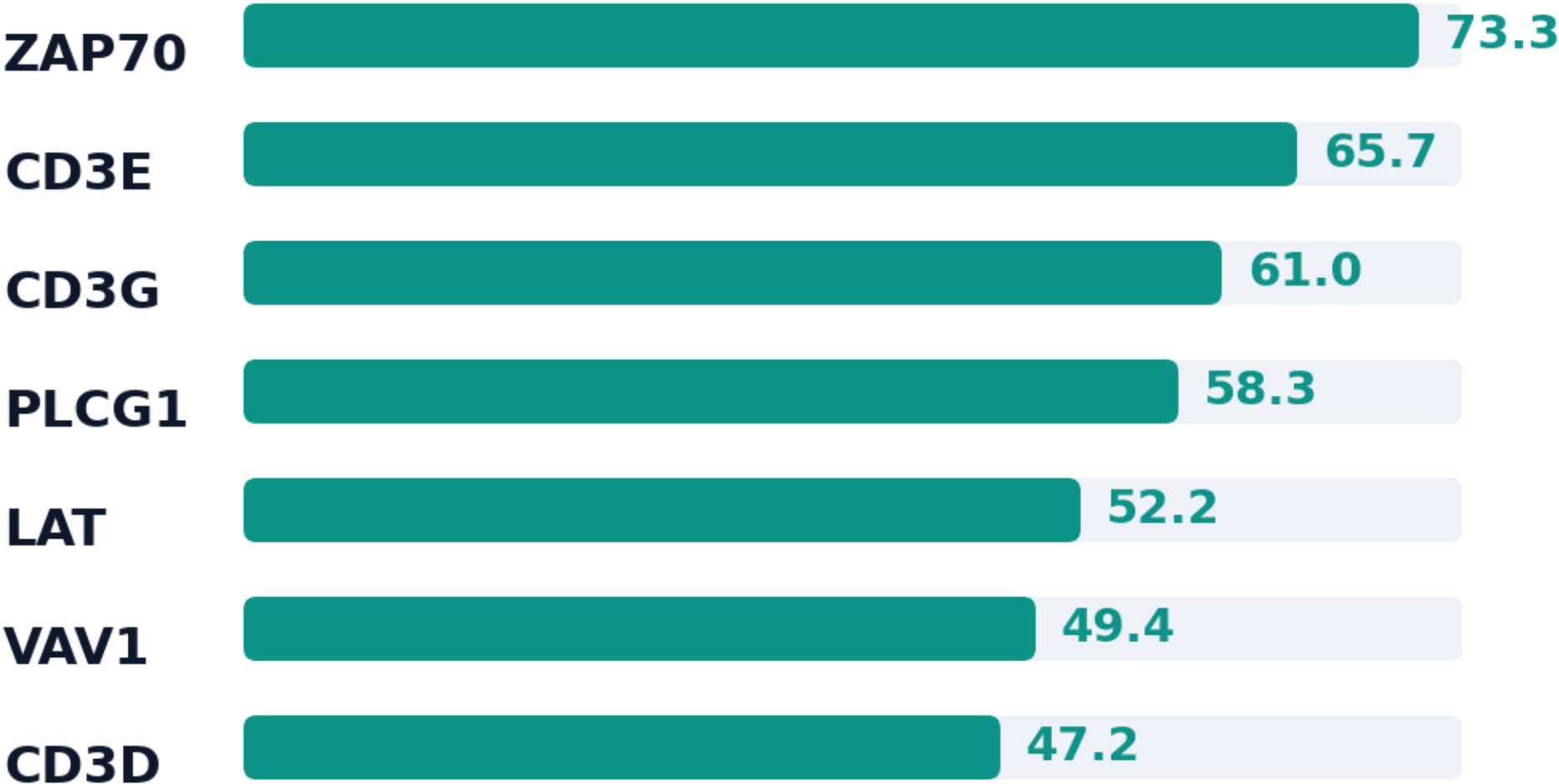
- The same aggregation yields donor-aware pseudobulk — and a signed direction-of-effect axis.

VALIDATION · GENOME-SCALE, UNSUPERVISED

First, it re-finds known biology

Unsupervised on 2.44M CD4⁺ T cells, the top causal hits are the entire TCR proximal-signaling module.

CAUSAL EFFECT SIZE (ENERGY DISTANCE)



RASA2

In the primary CD8⁺ arm, the same method independently re-nominates RASA2 — a CAR-T potency enhancer validated in Nature (2022) — from effect size alone.

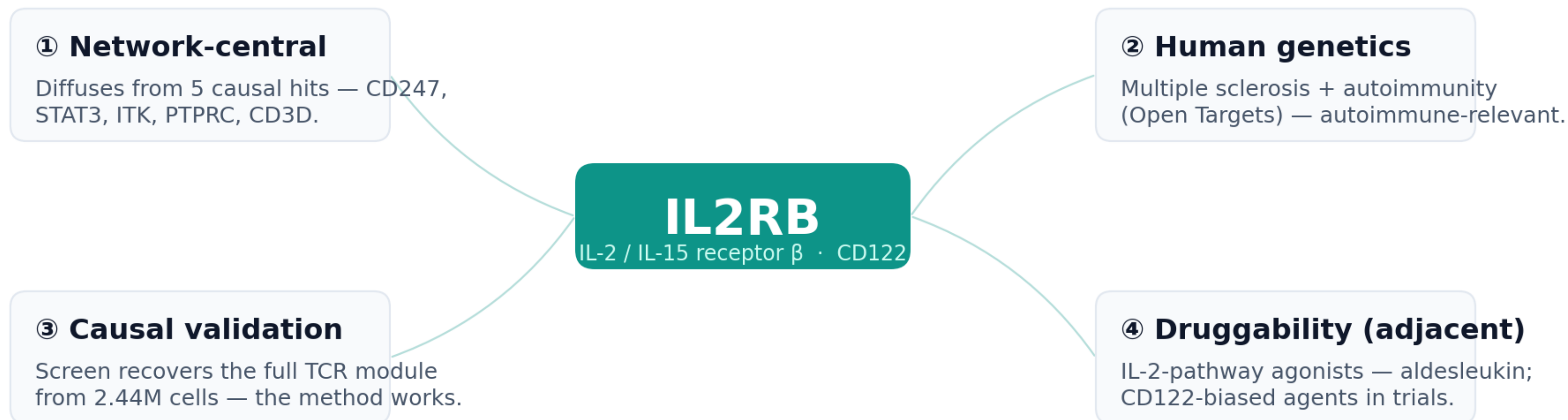
WHY IT MATTERS

If the method recovers what we know, we can trust what it finds next.

THE HERO · NETWORK-NOMINATED FROM THE GENOME-SCALE SCREEN

IL2RB — the IL-2 signaling axis

Never a direct hit — yet network-central among causal nodes.



VALIDATION

Top causal hits = the entire TCR module:
ZAP70, CD3D/E/G, PLCG1, LAT, VAV1, RASGRP1, ITK.

NOVEL FRONTIER

VAV2 · BLNK
genuinely undrugged.

OPEN SOURCE · ONE COMMAND

A real, actionable target from raw data — reproducibly.

Proposed validation: opposite-modality CRISPRa + an arrayed CRISPR readout in primary T cells, with a CD122-agonist counter-test for IL2RB.

DATA · CODE · FIGURES

All reproduce from one command.

REPRODUCIBLE PROVENANCE

Every claim carries its trail.

Built with Claude Science

Repo + landing page: [[REPO_URL](#)]